

Sales Forecasting & Demand Intelligence System

Executive Business Report

Executive Summary:

Historical retail sales show a clear long-term increase but also substantial month-to-month fluctuations, creating uncertainty in inventory and financial planning. The analysis compared three forecasting approaches, identified unusual sales events, forecast category and regional demand, and grouped products into four demand segments. XGBoost delivered the best overall forecasting accuracy with a 19.12% average percentage error, while SARIMA and Prophet provided useful forecast ranges for uncertainty planning. The East region showed the strongest upcoming growth at 131.11%, and the product analysis identified separate high-value, stable, high-volume, and rapidly growing demand groups. The main business implication is to use forecasts as a planning baseline, apply different stocking policies by demand segment, and investigate exceptional sales events before treating them as repeatable demand.

Key Findings from Sales Analysis and Forecasting:

The exploratory analysis found a **clear upward long-term sales trend**, with the smoothed monthly trend rising from approximately **40,000 to above 60,000** over the observed period. Sales also display **strong recurring seasonal movements** and substantial short-term volatility; first-order monthly changes frequently move by tens of thousands in either direction. This means that recent sales alone are not a reliable basis for purchasing decisions. Three models were evaluated on the **same final three-month test period** using MAE, RMSE, and MAPE. Lower values indicate better forecasting performance.

MODEL	MAE	RMSE	MAPE	ASSIGNMENT
XGBOOST	18710.45	20852.57	19.12	Best Overall
PROPHET	20296.01	22487.47	21.89	Strong Trend/Seasonality View
SARIMA	21060.32	22019.99	22.38	Useful Statistical Baseline

Finding: XGBoost achieved the lowest error across all three evaluation measures and is therefore the strongest model for the central short-term planning forecast. Prophet and SARIMA remain useful because they provide interpretable time-pattern analysis and uncertainty ranges.

Three-Month Sales Forecast and Planning Ranges:

MONTH	SARIMA FORECAST	SARIMA RANGE	PROPHET FORECAST	PROPHET RANGE	XGBOOST FORECAST
Jan 2019	48,057	21,353-74,762	42,548	33,076-52,195	44,980
Feb 2019	34,178	7,473-60,882	33,310	24,197-42,310	25,958
Mar 2019	68,800	42,095-95,504	80,305	71,330-89,677	54,966

Business interpretation: Both models indicate a weaker February followed by a strong March rebound. Because uncertainty remains wide, Supply Chain should use **staged replenishment**, while Finance should plan for **base, lower-demand, and higher-demand scenarios**.

Top Three Anomalies:

Z-Score and Isolation Forest did not identify exactly the same events because they detect abnormality differently. Priority events include:

DATE	SALES	DETECTION	LIKELY CAUSE
22 Mar 2015	37,703.67	Both Methods	Promotion,Festive Demand
2 Dec 2018	35,998.90	Isolation Forest	Year-end-Demand or Bulk Order
18 Nov 2018	30,572.45	Isolation Forest	Promotion or Festive Uplift

These causes are **business hypotheses and require validation** against promotion calendars, order records, and seasonal events. The 22 March 2015 event is especially important because both methods flagged it.

Product Demand Segmentation and Stocking Strategy:

Demand Segment	Key Finding	Stocking Strategy
High Value,Volatile	Sales 167,743;Highest Order Value 1,931	Moderate Safety Stock
Low Volume, Stable	Low Volatility,907	Lean Inventory
High Volume,Established	Highest Value,239,496	High Availability
Rapid Growth	Growth 192.84%	Gradual Stock Increase

Finding: One inventory policy should not be applied to all products. High-volume products need availability protection, stable low-volume products can operate leaner, volatile products need flexibility, and rapid-growth products require controlled expansion.

Three Priority Business Recommendations:

1. Use the best-performing forecast as the primary planning baseline.

XGBoost recorded the lowest average percentage error at **19.12%**, compared with **21.89%** and **22.38%** for the alternatives. Use it for short-term planning while retaining forecast ranges for uncertainty.

2. Shift inventory toward verified growth areas.

The **East region shows 131.11% upcoming growth**, while the rapid-growth product segment records **192.84% growth**. Increase inventory gradually in these areas and review performance frequently to limit overstock risk.

3. Create a formal exception-review process.

The **22 March 2015 sales spike of 37,704** was flagged by both anomaly approaches. Unusual peaks should be checked against promotions, bulk orders, and seasonal events before being included in future stocking assumptions.

Key Risk and Limitation:

The system relies on approximately **four years of historical sales data** and does not directly include important business drivers such as promotions, pricing changes, holidays, stock-outs, competitor activity, or economic conditions. Even the best approach has a **19.12% average percentage error**. Therefore, forecasts should support—not replace—management judgment, and performance should be reviewed and updated as new data becomes available.